Jie Ding

dingj@umn.edu www.jding.org

POSITION

University of Minnesota Minneapolis, MN
Associate Professor in Statistics 2023 - Present
Graduate Faculty in Electrical Engineering
Graduate Faculty in Computer Science
Core Member of the Data Science Initiative

Amazon Inc. Remote
Amazon Scholar 2022 - 2025

University of Minnesota Minneapolis, MN
Tenure-Track Assistant Professor in Statistics 2018 - 2023

Duke University Durham, NC
Postdoctoral Fellow (with Dr. Robert Calderbank) 2018

Harvard University Cambridge, MA
Postdoctoral Fellow (with Dr. Vahid Tarokh) 2017

EDUCATION

Harvard University Cambridge, MA
Ph.D., Engineering Sciences & A.M., Statistics 2012 - 2016
Ph.D. Thesis: Nonlinear Modeling and Prediction for Time Series

Tsinghua University Beijing, China
B.S., Mathematics (2008-2010) and Electrical Engineering (2010-2012) 2008-2012

FACULTY RECOGNITIONS

NSF CAREER Award 2024
Army Early Career Program (Young Investigator) Award 2023
Cisco Research Award - four times 2022-2025
AWS Cloud Credits for Research - two times 2021, 2022

Meta/Facebook Faculty Research Award - two times 2021, 2022

UMN Thank-A-Teacher Teaching Award – two times 2019, 2020

PUBLICATION

AI Foundations

- X. Wang, Q. Le, A. Ahmed, E. Diao, Y. Zhou, N. Baracaldo, J. Ding, A. Anwar, "MAP: Multi-Human-Value Alignment Palette," ***International Conference on Learning Representations (ICLR)***, **Oral Presentation** (1%), 2025.
- S. Moushegian, S. Wu, E. Diao, J. Ding, T. Banerjee, and V. Tarokh, "Robust Score-Based Quickest Change Detection," ***IEEE Transactions on Information Theory***, vol. 71, no. 7, pp. 5539-5555, 2025.
- J. Zhang, Y. Yang, J. Ding, "Information Criteria for Model Selection," ***WIREs Computational Statistics (invited article)***, 2023.
- G. Li, G. Wang, J. Ding, "Provable Identifiability of Two-Layer ReLU Neural Networks via LASSO Regularization," ***IEEE Transactions on Information Theory***, vol. 69, no. 9, pp. 5921-5935, 2023.
- G. Li, J. Ding, "Towards Understanding Variation-Constrained Deep Neural Networks," ***IEEE Transactions on Signal Processing***, vol. 71, pp. 631-640, 2023.
- C. Morchdi, Y. Zhou, J. Ding, B. Wang, "Exploring Gradient Oscillation in Deep Neural Network Training," ***59th Annual Allerton Conference on Communication, Control, and Computing (Allerton)***, 2023.
- S. Wu, E. Diao, T. Banerjee, J. Ding, V. Tarokh, "Quickest change detection for unnormalized statistical models," ***IEEE Transactions on Information Theory***, vol. 70, no. 2, pp. 1220-1232, 2024.
- J. Zhang, J. Ding, Y. Yang, "Is a Classification Procedure Good Enough? -- A Goodness-of-Fit Assessment Tool for Classification Learning," ***Journal of the American Statistical Association***, vol. 118, no. 542, pp. 1115-1125, 2023.
- J. Zhang, J. Ding, Y. Yang, "Targeted Cross-Validation," ***Bernoulli***, vol. 29, no. 1, pp. 377-402, 2023.
- J. Ding, E. Diao, J. Zhou, V. Tarokh, "On Statistical Efficiency in Learning," ***IEEE Transactions on Information Theory***, vol. 67, no. 4, pp. 2488-2506, 2021.
- G. Li, Y. Gu, J. Ding, "L1 Regularization in Two-Layer Neural Networks," ***IEEE Signal Processing Letters***, vol. 29, pp. 135-139, 2021.
- Y. Xiang, J. Ding, V. Tarokh, "Evolutionary Spectra Based on the Multitaper Method with Application to Stationarity Test," ***IEEE Transactions on Signal Processing***, vol. 67, no. 9, pp. 1353-1365, 2019.
- J. Ding, V. Tarokh and Y. Yang, "Bridging AIC and BIC: A New Criterion for Autoregression," ***IEEE Transactions on Information Theory***, vol. 64, no. 6, pp. 4024-4043, 2018.
- J. Ding, Y. Yang, V. Tarokh, "Model Selection Techniques--an overview," ***IEEE Signal Processing Magazine*** **(featured article)**, vol. 35, no. 6, pp. 16-34, 2018.
- J. Ding, S. Shahrampour, K. Heal, and V. Tarokh, "Analysis and Prediction of Multi-State Autoregressive

Models," ***IEEE Transactions on Signal Processing***, vol. 66, no. 9, pp. 2429-2440, 2018.

- J. Ding, Y. Xiang, L. Shen, and V. Tarokh, "Multiple Change Point Analysis: Fast Implementation and Strong Consistency," ***IEEE Transactions on Signal Processing***, vol. 65, no. 17, pp. 4495-4510, 2017.

Scalable Model Training and Inference

- Q. Le, E. Diao, Z. Wang, X. Wang, J. Ding, L. Yang, A. Anwar, "Probe Pruning: Accelerating LLMs through Dynamic Pruning via Model-Probing," ***International Conference on Learning Representations (ICLR)***, 2025.
- X. Xu, A. Hasan, J. Ding, V. Tarokh, "Meta-Learning Parabolic Partial Differential Equations," ***Conference on Uncertainty in Artificial Intelligence (UAI)***, 2024.
- X. Tang, J. Zhang, Y. He, X. Zhang, Z. Lin, S. Partarrieu, E. Hanna, Z. Ren, H. Shen, Y. Yang, X. Wang, N. Li, J. Ding, J. Liu. "Explainable Multi-Task Learning for Multi-Modality Biological Data Analysis." ***Nature Communications*** (Editors' Highlight), vol. 14, no. 2546, 2023.
- E. Diao, G. Wang, J. Zhang, Y. Yang, J. Ding, V. Tarokh, "Pruning Deep Neural Networks from a Sparsity Perspective," ***International Conference on Learning Representations (ICLR)***, 2023.
- E. Mushtaq, C. He, J. Ding, S. Avestimehr, "Distributed Architecture Search Over Heterogeneous Distributions," ***Transactions on Machine Learning Research (TMLR)***, 2023.
- S. Wu, E. Diao, J. Ding, T. Banerjee, V. Tarokh, "Robust Quickest Change Detection for Unnormalized Models," ***Conference on Uncertainty in Artificial Intelligence (UAI)***, 2023.
- C. Chen, J. Zhou, J. Ding, Y. Zhou, "Assisted Learning for Organizations with Limited Imbalanced Data," ***Transactions on Machine Learning Research***, 2023.
- X. Wang, J. Zhang, M. Hong, Y. Yang, J. Ding, "Parallel Assisted Learning," ***IEEE Transactions on Signal Processing***, vol. 70, pp. 5848-5858, 2022.
- E. Diao, J. Ding, V. Tarokh, "Gradient Assisted Learning," ***Conference on Neural Information Processing Systems (NeurIPS)***, 2022.
- M. Soltani, S. Wu, Y. Li, R. Ravier, J. Ding, and V. Tarokh, "Compressing Deep Networks Using Fisher Score of Feature Maps," ***Data Compression Conference (DCC)***, 2021.
- X. Xian, X. Wang, J. Ding, R. Ghanadan, "Assisted Learning: A Framework for Multi-Organization Learning," ***Conference on Neural Information Processing Systems (NeurIPS)***, **Spotlight Presentation** (3%), 2020.
- S. Shao, P. Jacob, J. Ding, V. Tarokh, "Bayesian Model Comparison with the Hyvarinen Score: Computation and Consistency," ***Journal of the American Statistical Association***, vol. 114, no. 528, pp. 1826-1837, 2019.
- K. Elkhalil, A. Hasan, J. Ding, S. Farsiu, V. Tarokh, "Fisher Auto-Encoders," ***International Conference on Artificial Intelligence and Statistics (AISTAT)***, 352-360, 2021.
- T. Xie, J. Ding, "Forecasting with Multiple Seasonality," ***IEEE International Conference on Big Data***

(***IEEE BigData***), 2020.

- S. Wu, E. Diao, J. Ding, V. Tarokh, "Deep Clustering of Compressed Variational Embeddings," ***Data Compression Conference (DCC)***, 2020.
- E. Diao, J. Ding, V. Tarokh, "DRASIC: Distributed Recurrent Autoencoder for Scalable Image Compression," ***Data Compression Conference (DCC)***, 2020.
- J. Ding, J. Zhou, and V. Tarokh, "Asymptotically Optimal Prediction for Time-Varying Data Generating Processes," ***IEEE Transactions on Information Theory***, vol. 65, no. 5, pp. 3034-3067, 2019.
- J. Ding, R. Calderbank, V. Tarokh, "Gradient Information for Representation and Modeling," ***Conference on Neural Information Processing Systems (NeurIPS)***, 2019.
- E. Diao, J. Ding, V. Tarokh, "Restricted Recurrent Neural Networks," ***IEEE International Conference on Big Data (IEEE BigData)***, 2019.
- S. Shahrampour, M. Noshad, J. Ding, V. Tarokh, "Online Learning for Multimodal Data Fusion with Application to Object Recognition," ***IEEE Transactions on Circuits and Systems II: Express Briefs***, vol. 65, no. 9, pp. 1259-1263, 2018.
- Q. Han, J. Ding, E. Airoldi, and V. Tarokh, "SLANTS: Sequential Adaptive Nonlinear Modeling of Time Series," ***IEEE Transactions on Signal Processing***, vol. 65, no. 19, pp. 4994-5005, 2017.

AI Safety and Privacy

- X. Wang, J. Du, A.A. Khan, Q. Le, E. Diao, J. Zhou, J. Ding, A. Anwar, "Beyond Expectations: Quantile-Guided Alignment for Risk-Calibrated Language Models," ***Conference on Neural Information Processing Systems (NeurIPS)***, **Spotlight**, 2025.
- G. Wang, X. Xian, X. Bi, J. Srinivasa, A. Kundu, C. Fleming, R. Zhang, M. Hong, J. Ding, "On the Vulnerability of Retrieval-Augmented Generation within Knowledge-Intensive Application Domains," ***International Conference on Machine Learning (ICML)***, 2025.
- X. Wang, E. Diao, Q. Le, J. Ding, A. Anwar, "AID: Adaptive Integration of Detectors for Safe AI with Large Language Models," ***Annual Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)***, 2025.
- G. Wang, X. Xian, X. Bi, J. Srinivasa, A. Kundu, M. Hong, J. Ding, "RAW: A Robust and Agile Plug-and-Play Watermark Framework for AI-Generated Images with Provable Guarantees," ***Conference on Neural Information Processing Systems (NeurIPS)***, 2024.
- G. Wang, X. Xian, X. Bi, J. Srinivasa, A. Kundu, M. Hong, J. Ding, "Demystifying Poisoning Backdoor Attacks from a Statistical Perspective," ***International Conference on Learning Representations (ICLR)***, 2024.
- X. Xian, G. Wang, J. Srinivasa, A. Kundu, X. Bi, M. Hong, J. Ding, "A Unified Framework for Inference-Stage Backdoor Defenses," ***Conference on Neural Information Processing Systems (NeurIPS)***, 2023.

- X. Xian, G. Wang, J. Srinivasa, A. Kundu, X. Bi, M. Hong, J. Ding, "Understanding Backdoor Attacks through the Adaptability Hypothesis," ***International Conference on Machine Learning (ICML)***, 2023.
- G. Wang, J. Ding, Y. Yang, "Regression with Set-Valued Categorical Predictors," ***Statistica Sinica***, vol. 33, no. 1, pp. 1-16, 2023.
- X. Xian, M. Hong, J. Ding, "Understanding Model Extraction Games," ***IEEE International Conference on Trust, Privacy and Security in Intelligent Systems, and Applications (TPS-ISA)***, 2022.
- J. Ding, B. Ding, "Interval Privacy: A Framework for Privacy-Preserving Data Collection," ***IEEE Transactions on Signal Processing***, vol. 70, pp. 2443-2459, 2022.
- X. Xian, M. Hong, J. Ding, "Mismatched Supervised Learning," ***International Conference on Acoustics, Speech, and Signal Processing (ICASSP)***, 2022.
- X. Wang, Y. Xiang, J. Gao, J. Ding, "Information Laundering for Model Privacy," ***International Conference on Learning Representations (ICLR)***, **Spotlight Presentation** (5.6%), 2021.
- J. Ding, B. Ding, "'To Tell You the Truth' by Interval-Private Data," ***IEEE International Conference on Big Data (IEEE BigData)***, 2020.

Agentic and Collaborative AI

- J. Zhang, Y. Yang, J. Ding, "Additive-Effect Assisted Learning," ***Journal of the Royal Statistical Society Series B: Statistical Methodology***, 2025.
- A. Luo, X. Xian, J. Du, F. Tian, G. Wang, M. Zhong, S. Zhao, X. Bi, Z. Liu, J. Zhou, J. Srinivasa, A. Kundu, C. Fleming, M. Hong, J. Ding, "AssistedDS: Benchmarking How External Domain Knowledge Assists LLMs in Automated Data Science," ***Conference on Empirical Methods in Natural Language Processing (EMNLP)***, 2025.
- Q. Le, E. Diao, X. Wang, V. Tarokh, J. Ding, A. Anwar, "DynamicFL: Dynamic Federated Learning with Communication Resource Allocation," ***IEEE International Conference on Big Data (BigData)***, ***Best Student Paper Award***, 2024.
- X. Wang, Q. Le, A. Khan, J. Ding, A. Anwar, "ICL: An Incentivized Collaborative Learning Framework," ***IEEE International Conference on Big Data (BigData)***, 2024.
- C. Chen, J. Zhang, J. Ding, Y. Zhou, "Assisted Unsupervised Domain Adaptation," ***IEEE International Symposium on Information Theory (ISIT)***, 2023.
- Q. Le, E. Diao, X. Wang, A. Anwar, V. Tarokh, J. Ding, "Personalized Federated Recommender Systems with Private and Partially Federated AutoEncoders," ***Asilomar Conference on Signals, Systems, and Computers (Asilomar)***, 2023.
- C. Ye, R. Ghanadan, J. Ding, "Meta Clustering for Collaborative Learning," ***Journal of Computational and Graphical Statistics***, vol. 32, no. 3, pp. 1160-1169, 2023.
- E. Diao, J. Ding, V. Tarokh, "SemiFL: Communication Efficient Semi-Supervised Federated Learning with Unlabeled Clients," ***Conference on Neural Information Processing Systems (NeurIPS)***, 2022.

- H. Chen, J. Ding, E. Tramel, S. Wu, A. Sahu, S. Avestimehr, T. Zhang, "Self-Aware Personalized Federated Learning," ***Conference on Neural Information Processing Systems (NeurIPS)***, 2022.
- E. Mushtaq, J. Ding, S. Avestimehr, "What If Kidney Tumor Segmentation Challenge (KiTS19) Never Happened," ***IEEE International Conference on Machine Learning and Applications (ICMLA)***, 2022.
- L. Collins, E. Diao, T. Roosta, J. Ding, T. Zhang, "PerFedSI: A Framework for Personalized Federated Learning with Side Information," ***International Workshop on Federated Learning: Recent Advances and New Challenges*** *in Conjunction with NeurIPS* ***(FL-NeurIPS'22)***, 2022.
- E. Diao, J. Ding, V. Tarokh, "HeteroFL: Computation and Communication Efficient Federated Learning for Heterogeneous Clients," ***International Conference on Learning Representations (ICLR)***, 2021.
- J. Zhou, J. Ding, K. M. Tan, and V. Tarokh, "Model Linkage Selection for Cooperative Learning," ***Journal of Machine Learning Research***, vol. 22, no. 256, pp. 1−44, 2021.

Miscellaneous

- J. Gao, J. Ding, "Large Deviation Principle for the Whittaker 2D Growth Model," arXiv preprint arXiv:2009.12907.
- J. Wang, M. Xue, and R. Culhane, E. Diao, J. Ding, and V. Tarokh, "Speech Emotion Recognition with Dual-Sequence LSTM Architecture," ***IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)***, 2020.
- J. Ding, M. Noshad, and V. Tarokh, "Complementary Lattice Arrays for Coded Aperture Imaging," ***Journal of the Optical Society of America***, vol. 33, no. 5, pp. 863-881, 2016.
- J. Ding, A. Bouabdallah, and V. Tarokh, "Key Pre-Distributions from Graph-Based Block Designs," ***IEEE Sensors Journal***, vol. 16, no. 6, pp. 1842-1850, 2015.
- J. Ding, L. Chen, Y. Gu, "Perturbation Analysis of Orthogonal Matching Pursuit," ***IEEE Transactions on Signal Processing***, vol. 61, no. 2, pp. 398-410, 2013.

MANUSCRIPTS UNDER REVIEW

- J. Du, X. Zhang, H. Shen, X. Xian, G. Wang, J. Zhang, Y. Yang, N. Li, J. Liu, J. Ding, "Drift to Remember," arXiv preprint arXiv:2409.13997.
- J. Du, Y. Yang, J. Ding, "Adaptive Continual Learning: Rapid Adaptation and Knowledge Refinement," 2023. Downloadable at https://jding.org/jie-uploads/2023/05/AdaptiveCL.pdf
- J. Du, Y. Yang, J. Ding, "Minimax Rate for Nonparametric Lifelong Learning," 2023.
- G. Wang, Y. Yang, J. Ding, "Model Privacy: A Unified Framework to Understand Model Stealing Attack and Defense," 2023.
- W. Yang, G. Wang, J. Ding, Y. Yang, "Theory-Guided Adaptive Backward Pruning of Deep Networks via Feature Retaining Index," 2023.

RESEARCH AWARDS

NSF CAREER (Ding)

Continual Learning with Evolving Memory, Soft Supervision, and Cross-Domain Knowledge - Foundational Theory and Advanced Algorithms 2024-29

Role: PI

Amount: $544,382

ARO Early Career Program, formerly known as the Young Investigator Program (Ding)

Information Fusion from Heterogeneous Sources 2023-26

Role: PI

Funded Amount: $351,916

NIH R01 (Liu/Li/Ding)

SCH: Al-driven Flexible Electronics for Cardiac Organoid Maturation 2023-27

Role: Multi-PI

Funded Amount: $1,200,000

NSF IMR (Zou)

MM-1A: Evolutionary Modeling and Acquisition of Multidimensional 5G Internet Measurements 2022-25

Role: co-PI

Funded Amount: $588,021

Cisco (Ding/Hong/Bi)

Understanding and Enhancing Safe Use of Multi-Agent Systems 2025-26

Role: PI

Funded Amount: $100,000

NSF Collaborative SCALE MoDL (Zhou/Ding) [Completed]

Advancing Theoretical Minimax Deep Learning: Optimization, Resilience, and Interpretability 2021-24

Role: PI

Funded Amount: $850,000

ONR core (Ding) [Completed]

Deep Adaptive Neural Structures 2021-24

Role: PI

Funded Amount: $705,010

Cisco (Ding/Hong/Bi) [Completed]

Efficient and Reliable Watermarking for Generative AI 2024-25

Role: PI

Funded Amount: $139,869

Cisco (Ding/Hong/Bi) [Completed]
New Foundations and Algorithms for Advancing Adversarial Learning 2023-24
Role: PI
Funded Amount: $155,664

Cisco (Ding/Hong/Bi) [Completed]
Privacy-Preserving Machine Learning and Decision Making 2022-23
Role: PI
Funded Amount: $150,388

NSF Collaborative CPS Medium (Liu) [Completed]
An AI-enabled Cyber-Physical-Biological System for Cardiac Organoid Maturation 2020-23
Role: co-PI
Funded Amount: $898,225

ARO core (Ding) [Completed]
Robust Methods for Trustworthy Machine Learning 2020-23
Role: PI
Funded Amount: $355,202

UMN Grant-in-Aid (Ding) [Completed]
Dynamical Modeling of Complex Data 2020-22
Role: PI
Funded Amount: $47,814

DARPA (Tarokh) [Completed]
Physics for Artificial Intelligence 2018-20
Role: co-PI
Funded Amount: $488,065

TEACHING EXPERIENCE

Fall 2024: STAT 8901 - Generative AI: Principles and Practices (**Newly Created Course**)

Spring 2024: STAT/CSCI/AST/PHYS 8581 - Big Data in Astrophysics

Spring 2023: STAT/CSCI/AST/PHYS 8581 - Big Data in Astrophysics

Spring 2022: STAT 4052 - Intro to Statistical Learning

Spring 2022: STAT 8112 - Mathematical Statistics II

Spring 2021: STAT 8112 - Mathematical Statistics II

Fall 2020: STAT 5302 - Applied Regression Analysis

Spring 2020: STAT 8112 - Mathematical Statistics II

Fall 2019: STAT 5302 - Applied Regression Analysis

Spring 2019: STAT 5021 - Statistical Analysis

Fall 2018: STAT 5302 - Applied Regression Analysis

PATENTS

J. Ding, H. Chen, T. Zhang, "Updating Machine Learning Models Across Devices," 02/22/2022, US patent, #17/677,614.

J. Ding, X. Xian, X. Wang, "Assisted Learning and Module Privacy," 02/12/2020, US patent, #62975348.

J. Ding, W. Hua, "Single-Epoch Pseudo-Range Positioning Under Varying Ionosphere Delays," 06/05/2020, US patent #16894492.

J. Ding, W. Hua, "Rapid Determination of An Unknown Position," 03/08/2018, US patent #151915974.

INVITED TALKS

- "Advancing Scalable AI: From Core Principles to Modern Applications," invited talk at the University of Minnesota Data Science Initiative Workshop on AI, May 2025.
- "Heterogeneous Supervision of Autonomous Agents in Networked AI," invited seminar talk at Duke University, Mar 21, 2023.
- "Scalable Training and Security Maintenance for Large AI Models," invited streaming talk at Cisco Research Generative AI and Security Summit, virtual, October 3, 2023.
- "A Unified Framework for Understanding and Quantifying Model Privacy in Adversarial Machine Learning," invited session "The Art and Science of Predictive Modeling: from Theory to Practice" at EcoSta2023, Tokyo, Japan, August 2, 2023.
- "Pruning Deep Neural Networks from A Sparsity Perspective," invited session "Trustworthy Machine Learning Methods and Applications" at EcoSta2023, Tokyo, Japan, August 1, 2023.
- "Model Privacy: A Unified Framework to Understand Model Stealing Attack and Defense," invited talk at 2023 North America Machine Learning, Optimization, and Statistics Symposium (NAMOS), Vancouver, Canada, June 24, 2023.
- "Interval Privacy: A New Framework for Privacy-Preserving Data Collection," ENAR invited session on "Analysis of Distributed Health Data: Novel Approaches and Applications", Nashville, Tennessee, March 19, 2023.
- "Interval Privacy: A New Framework for Privacy-Preserving Data Collection," CMStatistics invited session on "Federate Learning and Data Privacy in Modern Data Analysis", London, UK, December 17, 2022.
- "Privacy-Preserving Multi-Target Multi-Domain Recommender Systems," Production and Operations Management Society Annual Conference (POMS), April 21, 2022.

- "Human-Centric Privacy-Preserving Data Collection via Intervals," Department of Applied Economics and Statistics, University of Delaware, Mar. 11, 2022.
- "Interval Privacy: A New Framework for Privacy-Preserving Data Collection," 56th Annual Conference on Information Sciences and Systems (CISS), Mar. 9, 2022.
- "Interval Privacy: A New Framework for Privacy-Preserving Data Collection," Department of Statistics and Actuarial Science, University of Iowa, Feb. 10, 2022.
- "Organizational Collaboration with Assisted Learning," Computer Science Department Colloquium, Texas Tech University, Nov. 9, 2021.
- "Organizational Collaboration with Assisted Learning," IMA Data Science Seminar, Oct. 12, 2021.
- "Privacy-Sensitive Collaborations through Assisted Learning," Department of Statistics, University of Virginia, Feb. 26, 2021.
- "Privacy-Sensitive Collaborations through Assisted Learning," Department of Electrical and Computer Engineering, University of Utah, Feb. 19, 2021.
- "Privacy-Sensitive Machine Learning," Alexa AI, Amazon, Jan. 7, 2021.
- "Kinetic Prediction -- Predicting Time Series with Abrupt Changes and Smooth Evolutions," School of Finance and Statistics, East China Normal University, Dec. 18, 2019.
- "Modeling, Prediction, and Diagnostics for Trustworthy AI," Interdisciplinary Distinguished Seminar Series (IDSS), Department of Electrical and Computer Engineering, North Carolina State University, Oct. 25, 2019.
- "Model Selection Principles for Data Analysis," DTC Seminar Series, Digital Technology Center, University of Minnesota, Sept. 30, 2019.
- "Modeling, Prediction, and Diagnostics from Online Streaming Data," ECE Colloquium Series, Department of Electrical and Computer Engineering, University of Minnesota, Sept. 26, 2019.
- "Predicting Time Series with Abrupt Changes and Smooth Evolutions," International Conference on Econometrics and Statistics (EcoSta), June 26, 2019.

SERVICES

Associate Editor, IEEE Transactions on Information Theory June 2025 - present

Area Chair, ICLR-26 Aug 2025

Program Committee, AAAI-26 July 2025

Elected Member, Nomination & Election Subcommittee Chair (in 2020)
Machine Learning for Signal Processing Technical Committee, IEEE Nov. 2018 - Dec. 2024

Local Organizer Committee*, Midwest Machine Learning Symposium (MMLS 2024)* May 20-21, 2024

Co-Organizer, *UMN Machine Learning Seminar Series* Jan. 2021 - present

Co-Organizer, *IEEE International Conference on Acoustics, Speech and Signal Processing, special session on*

"Federated Learning Challenges and Opportunities: An Outlook" May 2022

Program Committee, *International Workshop on "Federated Learning for User Privacy and Data Confidentiality" in Conjunction with ICML 2021* June 2021

Program Committee, *IEEE International Conference on Big Data* Dec. 2021

Co-Organizer, *IEEE BigData Workshop* on "Scalable Reinforcement Learning with Big Data" Dec. 2021

Faculty Lead, *Data+ outreach program* May 2019 - Aug. 2019

Area Chair, *IEEE International Conference on Acoustics, Speech and Signal Processing* May 2020

Committee Member Sept. 2018 - present

of Master and Ph.D. Committees from the School of Statistics, Department of Electrical and Computer Engineering, Department of Computer Science, Department of Psychology, College of Design, and Department of Agriculture at the University of Minnesota, Twin Cities

Reviewer of

NeurIPS, ICLR, Journal of Machine Learning Research, Annals of Statistics, Journal of the American Statistical Association, IEEE Transactions on Signal Processing, IEEE Transactions on Information Theory, etc.

ACADEMIC & COMMITTEE ADVISING

Doctoral Advisees

Jiawei Zhang, STAT Ph.D. (Co-advised with Dr. Yuhong Yang)	2018 – 2023
(Currently Assistant Professor at University of Kentucky)	
Ganghua Wang, STAT Ph.D. (Co-advised with Dr. Yuhong Yang)	2020 – Present
(Currently Postdoc Research at University of Chicago)	
Xun Xian, ECE Ph.D. (Co-advised with Dr. Mingyi Hong)	2018 – Present
Qi Le, CS Ph.D. (Co-advised with Dr. Ali Anwar)	2022 - Present
Jiaying Zhou, STAT Ph.D. (Co-advised with Dr. Yuhong Yang)	2018 - Present
Jin Du, STAT Ph.D. (Co-advised with Dr. Yuhong Yang)	2022 – Present
An Luo, STAT Ph.D.	2023 – Present
Fangqiao Tian, Statistics Ph.D.	2024 – Present

Doctoral Preliminary Committee: Committee Chair

Mitchell Kinney, Statistics Ph.D.	2018 - 2020

Doctoral Final Committee: Committee Reviewer

Seonmo Kim, Computer Science Ph.D.	2021
Vasileios Georgios Karanikolas, Electrical Engineering Ph.D.	2021

Doctoral Preliminary Committee: Committee Member

Daniel Ngo, Computer Science Ph.D.	2020
Wenjing Yang, Statistics Ph.D.	2020

Rui Wang, Statistics Ph.D.	2020
Vasileios Georgios Karanikolas, Electrical Engineering Ph.D.	2019
Bingxin Zhao, Statistics Ph.D.	2019

Master's Thesis/Research Committee: Committee Chair

Tianyang Xie, Statistics M.S.	2020
Yilin Hou, Statistics M.S.	2019
Eunji Min, Statistics M.S.	2019
Yi Rong, Statistics M.S.	2019
Suya Wu, Statistics M.S.	2019
Mitchell Kinney, Statistics M.S.	2018

Master's Thesis/Research Committee: Committee Member

Rutvij Umesh Bora, Data Science M.S.	2021
Xiangyi Chen, Elec & Comp Eng M.S.	2021
Anubha Agrawal, Data Science M.S.	2021
Sheng Huang, Data Science M.S.	2021
Sai Kumar Kayala, Data Science M.S.	2021
Aditya Anirudha Gaydhani, Data Science M.S.	2021
Anushree Choudhary, Data Science M.S.	2021
Mourya Karan Reddy Baddam, Data Science M.S.	2021
Yuanyuan Qiu, Data Science M.S.	2021
Buu Huynh, Statistics M.S.	2021
Rachit Jas, Data Science M.S.	2020
Wenjing Yang, Statistics M.S.	2020
Zachary Brown, Biostatistics M.S.	2020
Rui Wang, Statistics M.S.	2020
Rabin KC, Applied Plant Sciences M.S.	2020
Anchit Sharma, Data Science M.S.	2020
Garim Lee, Design M.S.	2019 - 2020
Bingxin Zhao, Statistics M.S.	2019 - 2020
Lanhuizi Gan, Mass Communication M.A.	2019
Xiaochen Jin, Statistics M.S.	2019
Elise Anderson, Psychology M.A.	2019